CSC12107 – INFORMATION SYSTEMS FOR BUSINESS INTELLIGENCE

PRACTICAL PROJECT

1. General Information

Assignment Code:	2026_DATH
Expected Duration:	8 tuần
Submission Deadline:	12/2026
Format:	Bài tập Assignment
Submission Method:	Submit via Course Moodle
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2. Target Course Learning Outcomes (CLOs)

This assignment aims to achieve the following course learning outcomes:

- G3.2: Explain the role and characteristics of data warehouses, the significance of dimension and fact tables, attributes, and dimensional hierarchies in multidimensional schemas.
- G3.3: Design multidimensional schemas (star, snowflake) based on transactional operational data and analytical requirements from a given business scenario.
- G4.2: Distinguish the applications of data mining in business intelligence.
  Explain the characteristics and purposes of data mining, its importance, and several popular techniques.
- G5.1: Implement an ETL process to extract data from multiple sources, transform and clean data, and load data into the Data Warehouse.
- G5.2: Construct a multidimensional Data Warehouse and justify the selection of appropriate OLAP operations for given analytical requirements.
- G5.3: Utilize data visualization tools (PowerBI, Excel, etc.) to present analytical and mined results (reports, dashboards, etc.).

3. Project Description

The project dataset describes Yellow Taxi trip records in New York City (NYC) from 2009 to 2026:

- **Datasets:**
  - Yellow Taxi Trip Records Data
  - Taxi Zones Lookup (CSV)
  - Taxi Zones Boundary (Shapefile ZIP)
- **Analytical Requirements**
  - **Data Profiling & Survey: (5%)**
    1. Survey and describe data sources

2. Analyze raw data anomalies

- **Basic Requirements**: **(80%)**
  1. Statistics on the number of taxi trips and passenger count by day ➔ month ➔ year.
  2. Analyze revenue for each borough/zone (Manhattan, Brooklyn, Queens, Bronx, Staten Island) by each vendor (CMT, VTS) at a specific point in time (date, hour, month, year).
  3. Statistics on the number of trips by payment type (Cash, Credit card, Dispute, No Charge, etc.) in each borough/zone by month of the year.
  4. Statistics on the number of pickups and drop-offs by time period (time of day / day of week / month of year).
  5. Statistics on the number of trips by passenger count group (solo passenger, group of 2-4, group of 5 or more) across each rate code type and trip distance.
  6. Build a dashboard analyzing taxi transportation performance for each borough/zone:
  7. Top n most popular routes
  8. Total revenue
- **Advanced Analytical Requirements: (15%)**
  1. Compare the average tip percentage and average trip speed across each trip rate code type.
  2. Did May revenue increase or decrease compared to June? Why? Explain the underlying reasons.
  3. Revenue increased while the number of trips remained nearly unchanged. Which factors drove the increase in revenue per trip?
  4. Compare revenue and trip volume over the same period (current month vs. previous month / current year vs. previous year).
  5. Build a dashboard analyzing taxi transportation performance for each borough/zone:

  a. Display pickup and drop-off volume by geographic zone on a spatial map.
  b. Create a table displaying revenue by borough across months of the year; plot trend charts analyzing performance by time of day.

- **Design Requirements**
  - Design the Dimensional Data Store (DDS) multidimensional schema
  - Design a 3-tier Data Warehouse architecture: ***Source > Staging > DDS***
  - Design test suites (testing framework) hỗ trợ đỗ nhiều lần, ghi log lại, lưu scd
  - Design an automated ETL workflow pipeline
  - Design dashboard user interfaces
- **Implementation Requirements**
  - Implement Metadata
  - Execute data loading into the warehouse: ***Staging, Fact/Dims***
  - Testing & Validation

4. Detailed Submission Requirements & Regulations

- You may use Word, Excel, Draw.io, etc., to diagram the above schemas.
- The submission must clearly state the Group ID, Student IDs, and Full Names at the beginning, along with individual task assignments and performance evaluations for each member and the entire team.
- Naming format: **MaNhom_MSSV1_MSSV2_MSSV3_ MSSV4_ MSSV5**

5. **Evaluation & Grading Criteria**

No.	Assessment Content	Weight
1	Data profiling & survey analysis	5%
2	Requirements analysis & DDS design	15%
3	ETL implementation & data loading	30%
4	Report / OLAP / Dashboard	35%
5	Advanced requirements	15%
	Total	**100%**